Teorija polj

Vektorski operatorji

s = skalarno polje, V = vektorsko polje

grad: $\nabla s = \left(\frac{\partial s}{\partial x}, \frac{\partial s}{\partial y}, \frac{\partial s}{\partial z}\right)$

div: $\nabla \cdot \mathbf{V} = \frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z}$

rot: $\nabla \times \mathbf{V} = \begin{vmatrix} \mathbf{i} & \mathbf{j} & \mathbf{k} \\ \frac{\partial}{\partial x} & \frac{\partial}{\partial y} & \frac{\partial}{\partial z} \\ V_x & V_y & V_z \end{vmatrix} =$

$= (Py - Qz, Pz - Rx, Qx - Py)$

Laplacijan: $\nabla^2 s = \frac{\partial^2 s}{\partial x^2} + \frac{\partial^2 s}{\partial y^2} + \frac{\partial^2 s}{\partial z^2}$

vek. produkt treh vektorjev: BAC-CAB

$\nabla \times (\nabla \times \vec{v}) = \nabla(\nabla \cdot \vec{v}) - \vec{v}(\nabla \cdot \nabla))$

Identiteta dvokratnega rotorja:

$rot(rot\vec{v}) = grad(div\vec{v}) - \nabla^2\vec{v}$

Prostorska geometrija

Koordinatni sistemi

Polarni: $x = r\cos\theta, \quad y = r\sin\theta \quad J = r$

Cilindrični:

$x = r\cos\theta, \quad y = r\sin\theta, \quad z = z \quad J = r$

Sferični: $x = r\sin\phi\cos\theta, \quad y = r\sin\phi\sin\theta,$

$z = r\cos\phi \quad J = r^2\sin\phi$

$J = det\begin{bmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{bmatrix} = det\begin{bmatrix} x_u & x_v \\ y_u & y_v \end{bmatrix}$

Razno formule

Ločna dolžina: $s = \int_a^b |\mathbf{r}'(t)|\, dt =$

$= \int_a^b \sqrt{x'(t)^2 + y'(t)^2 + z'(t)^2}\, dt$

parame. točk: $\vec{r} = r_A * t(r_B - r_A)$

Odvod integrala s parametrom:

$F(x) = \int_{u(x)}^{v(x)} f(x,y)dy \quad \Rightarrow$

$F'(x) = \int_{u(x)}^{v(x)} \frac{\partial(f(x,y))}{\partial(x)}dy +$

$+ f(x, v(x)) * v'(x) - f(x, u(x)) * u'(x)$

Naravna parameterizacija:

$|\mathbf{r}'(s)|^2 = 1, \quad s =$ ločna dolžina

Tangentna premica in normalna ravnina:

$\vec{e}$ ali $\vec{n} = \dot{\vec{r}}(t0)$

Kot med Vektorjema: $\cos\theta = \frac{\vec{a} \cdot \vec{b}}{|\vec{a}||\vec{b}|}$

Večkratni integrali

Dvojni integral:

$\iint_D f(x,y)\, dA = \int_a^b \int_{g_1(x)}^{g_2(x)} f(x,y)\, dy\, dx$

Ploščina: $A = \iint_D 1\, dA$

Prostornina: $V = \iint_D f(x,y)\, dA$

Težišče: $\bar{x} = \frac{1}{A}\iint_D x\, dA, \quad \bar{y} = \frac{1}{A}\iint_D y\, dA$

Masa: $m = \iiint_E \rho(x,y,z)\, dV$

Vztrajnostni moment: $I = \iiint_E r^2\rho\, dV$

Krivuljni integrali

S. polje: $\int_C f(x,y,z)ds =$

$= \int_{t0}^{t1} f(x(t), y(t), z(t)) * \sqrt{\dot{x}^2, + \dot{y}^2 + \dot{z}^2}dt$

V. polje: $\int_C \mathbf{V} \cdot d\mathbf{r} = \int_C (Pdx + Qdy + Rdz) =$

$= \int_{t0}^{t1} (P\dot{x} + Q\dot{y} + R\dot{z})dt$

$d\vec{r} = \dot{\vec{r}}dt = (\dot{x}dt, \dot{y}dt, \dot{z}dt)$

Neodvisen od poti če $\vec{v}$ potencialno polje $(rot\vec{V} = 0)$

Ploskovni integrali

S. polje: $\iint_S f\, dS =$

$= \iint_D f(\mathbf{r}(u,v))\sqrt{EG - F^2}\, du\, dv$

kjer $E = \mathbf{r}_u \cdot \mathbf{r}_u, \quad F = \mathbf{r}_u \cdot \mathbf{r}_v, \quad G = \mathbf{r}_v \cdot \mathbf{r}_v$

$z = z(x,y), \quad \vec{n} = (-z_x, -z_y, 1)$

V. polje: $\iint_S \mathbf{V} \cdot d\mathbf{S} =$

$= \iint_S \mathbf{V} \cdot \vec{n}\, dS = \iint_D [\mathbf{V}, \mathbf{r}_u, \mathbf{r}_v]\, du\, dv$

$\vec{v} = \frac{\vec{r}_u \times \vec{r}_v}{|\vec{r}_u \times \vec{r}_v|}, \quad |\vec{r}_u \times \vec{r}_v| = \vec{n}$

Integralski izreki

Greenov, Stokesov, Gaussov :

$\oint_C (P\, dx + Q\, dy) = \iint_D \left(\frac{\partial Q}{\partial x} - \frac{\partial P}{\partial y}\right) dxdy$

C : zaključena pozitivno orientirana krivulja, ki je rob območja D

$\oint_C \mathbf{V} \cdot d\mathbf{r} = \iint_S (\nabla \times \mathbf{V}) \cdot d\mathbf{S}$

C : rob površine S

$\iint_S \mathbf{V} \cdot d\mathbf{S} = \iiint_E \nabla \cdot \mathbf{V}\, dV$

S : zaprt površina, ki je rob območja E

Kompleksna analiza

Elementarne komp. fun.

$z = x + iy; \quad |z| = \sqrt{x^2 + y^2};$

$\arg z = \arctan\left(\frac{y}{x}\right)$

$e^z = e^x(\cos y + i\sin y); \quad d\varphi = \frac{dz}{iz}$

$\log z = \log|z| + i(\arg z + 2\pi k)$

$\sin z = \frac{e^{iz} - e^{-iz}}{2i}, \cos z = \frac{e^{iz} + e^{-iz}}{2}$

$\sinh z = \frac{e^z - e^{-z}}{2}, \cosh z = \frac{e^z + e^{-z}}{2}$

Integral Kompleksne funkcije

$\int_C f(z)\, dz = \int_C (u + iv)(dx + idy) =$

$= \int_C (u\, dx - v\, dy) + i\int_C (v\, dx + u\, dy)$

$= \int_{t_a}^{t_b} f(z(t)) \cdot \dot{z}(t)\, dt$

Cauchjeva form.: $f(z_0) = \frac{1}{2\pi i}\oint_C \frac{f(z)}{z - z_0}dz$

!Integral analitične. fun. ni odvisen od poti!

Taylorjeva in Laurentova vrsta

$f(x) = \sum_{n=1}^{\infty} \frac{f^{(n)}(x_0)}{n!}(x - x_0)^n$

Geometrijska, sin, cos, e vrsta:

$\frac{1}{1 - z} = \sum_{n=0}^{\infty} z^n \quad ; |z| < 1$

$\sin z = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n+1}}{(2n+1)!} \quad$; za vse z

$\cos z = \sum_{n=0}^{\infty} (-1)^n \frac{z^{2n}}{(2n)!} \quad$; za vse z

$e^z = \sum_{n=0}^{\infty} \frac{z^n}{n!} \quad$; za vse z

Laurentova vrsta:

$f(z) = \sum_{n=-\infty}^{\infty} a_n(z - z_0)^n$

$a_n = \frac{1}{2\pi i}\oint_C \frac{f(z)}{(z - z_0)^{n+1}}dz$

Analitičnost in odvedljivost in harmoničnost

C-R pogoj:

$\frac{\partial u}{\partial x} = \frac{\partial v}{\partial y}, \quad \frac{\partial u}{\partial y} = -\frac{\partial v}{\partial x}$

$u = \int v_y\, dx = -\int v_x\, dy + C$

Pogoj harmoničnosti:

$\nabla^2 u = \nabla^2 v = 0 // w_{xx} + w_{yy} = 0$

Formule kompleksne analize

Residum: $\oint_C f(z)dz = 2\pi i \sum \text{Res}(f, z_k)$

$Resf(z) = \frac{1}{(m-1)!} *$

$\lim_{z \to z_0} [\frac{d^{m-1}}{dz^{m-1}}((z - z_0)^m f(z))]$

[m]-pol n-te stopnje

L.V.:$f(z) = \sum_{n=-\infty}^{\infty} C_n(z - z_0)^n$

$C_{-1} = Resf(z) \,||\, f'(z)$ za stopnje ničel

Eul. form. v $\mathbb{C}$: (realna rac. funkcija [f])

$sin(\phi) = \frac{e^{i\phi} - e^{-i\phi}}{2i}$

$cos(\phi) = \frac{e^{i\phi} + e^{-i\phi}}{2}$

$z = e^{i\phi}|d\phi = dz/iz| \int_0^{2\pi} f(sin(\phi), cos(\phi))d\phi$

$\frac{z - z^{-1}}{2i} = sin(\phi) \mid \frac{z + z^{-1}}{2} = cos(\phi)$

Konformne presl. (C-R), Lomljena Linearna presl. / Mobiusova presl.

Preslikava: $w = f(z) = u(x,y) + iv(x,y)$

$f(z) = \frac{az + b}{cz + d}; ad - bc \neq 0$

Integrali

Pravila integriranja

$$\text{Substitucija: } \int f(g(x))g'(x)dx = \int f(u)du, \quad u = g(x)$$

$$\text{Po delih: } \int u\,dv = uv - \int v\,du$$

$$\text{Parcialni ulomki: } \frac{P(x)}{Q(x)} = \sum \frac{A_i}{(x-a_i)^{n_i}}$$

Elementarni integrali

$$\int x^n dx = \frac{x^{n+1}}{n+1} + C, \quad n \neq -1$$

$$\int \frac{1}{x} dx = \ln|x| + C$$

$$\int e^x dx = e^x + C$$

$$\int \sin x dx = -\cos x + C$$

$$\int \cos x dx = \sin x + C$$

$$\int \frac{1}{1+x^2} dx = \arctan x + C$$

$$\int \frac{1}{\sqrt{1-x^2}} dx = \arcsin x + C$$

Odvodi

Pravila odvajanja

$$(f+g)' = f' + g'$$

$$(fg)' = f'g + fg'$$

$$\left(\frac{f}{g}\right)' = \frac{f'g - fg'}{g^2}$$

$$(f(g(x)))' = f'(g(x)) \cdot g'(x)$$

Elementarni odvodi

$$\frac{d}{dx} x^n = nx^{n-1}$$

$$\frac{d}{dx} e^x = e^x$$

$$\frac{d}{dx} \ln x = \frac{1}{x}$$

$$\frac{d}{dx} \sin x = \cos x$$

$$\frac{d}{dx} \cos x = -\sin x$$

$$\frac{d}{dx} \tan x = \sec^2 x$$

$$\frac{d}{dx} \arctan x = \frac{1}{1+x^2}$$

$$\frac{d}{dx} \arcsin x = \frac{1}{\sqrt{1-x^2}}$$

Trigonometrske funkcije

$$\sin^2 x + \cos^2 x = 1$$

$$\tan x = \frac{\sin x}{\cos x}$$

$$\sin 2x = 2\sin x \cos x$$

$$\cos 2x = \cos^2 x - \sin^2 x$$

$$\tan 2x = \frac{2\tan x}{1 - \tan^2 x}$$

$$\sin(x \pm y) = \sin x \cos y \pm \cos x \sin y$$

$$\cos(x \pm y) = \cos x \cos y \mp \sin x \sin y$$

$$\tan(x \pm y) = \frac{\tan x \pm \tan y}{1 \mp \tan x \tan y}$$

Adicijski izrek za sin,cos :

$$\sin(x+iy) = \sin x \cosh y + i \cos x \sinh y$$

$$\cos(x+iy) = \cos x \cosh y - i \sin x \sinh y$$